

SPARSE, LOW-RANK, AND DEEP: A COMPRESSED-SENSING FRAMEWORK FOR MUSIC SOURCE SEPARATION

Taka Khoo

Thayer School of Engineering, Dartmouth College · ENGS 109: High-Dimensional Sensing and Learning (Spring 2026)

The problem, and why it is personal

Music source separation splits a finished mix into stems (vocals, accompaniment). It is the suggested project bullet “*isolate instrumental music from vocal component*,” and it is the third time I have studied audio under Peter Chin:

I Undergrad thesis: a 1.08B-param Token U-Net for full-mix *restoration* on EnCodec tokens.

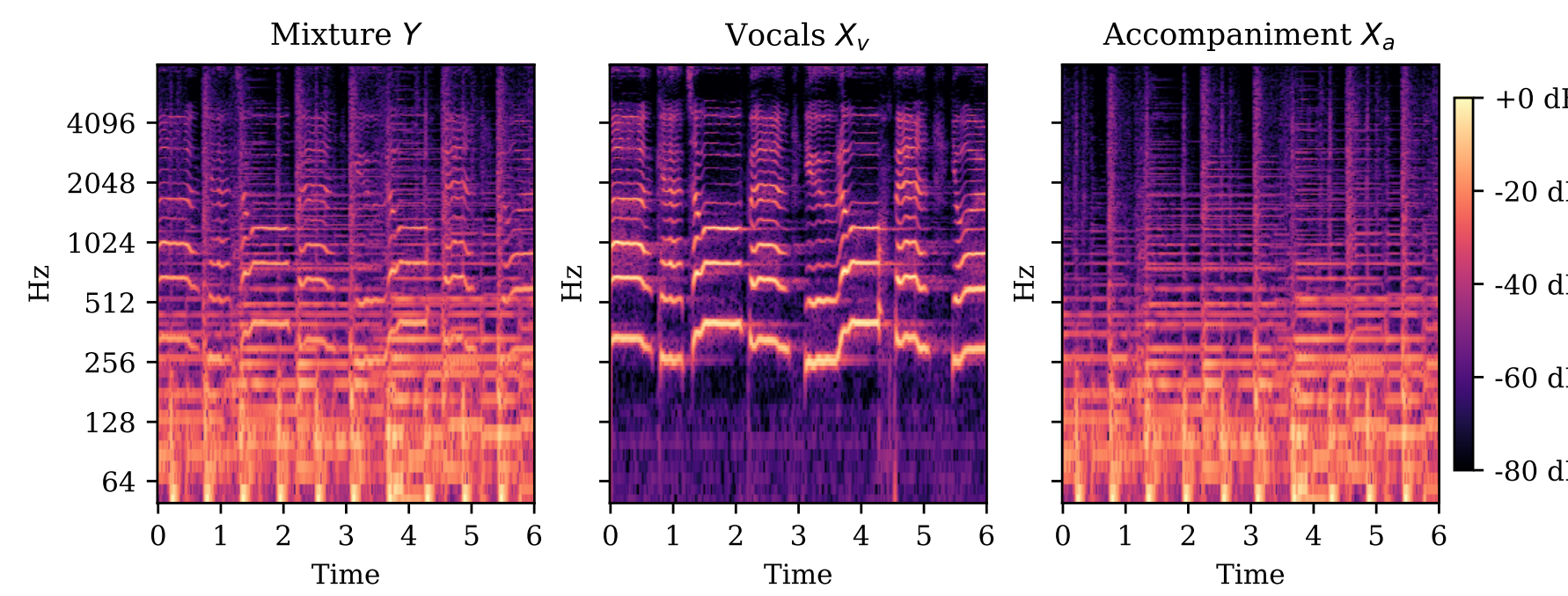
II MS thesis: MODULO / Mozart AI, a co-creative DAW whose stem *separation* is outsourced to ElevenLabs (≈ 35 dB SI-SDR below Demucs).

III This project: attack separation from the compressed-sensing angle, with the recovery theory the deep baselines lack.

Music separation *is* compressed sensing

On the STFT magnitude $\mathbf{M} = |\text{STFT}(y)|$:

- Vocals are **sparse in frequency** (thin formants); drums **sparse in time** (transients).
- Sustained harmony is **low-rank** (L13).
- Stereo channels share active sources: **joint sparsity** / **MMV** (L12).
- Deep nets work because they **unroll sparse coding** (L16).

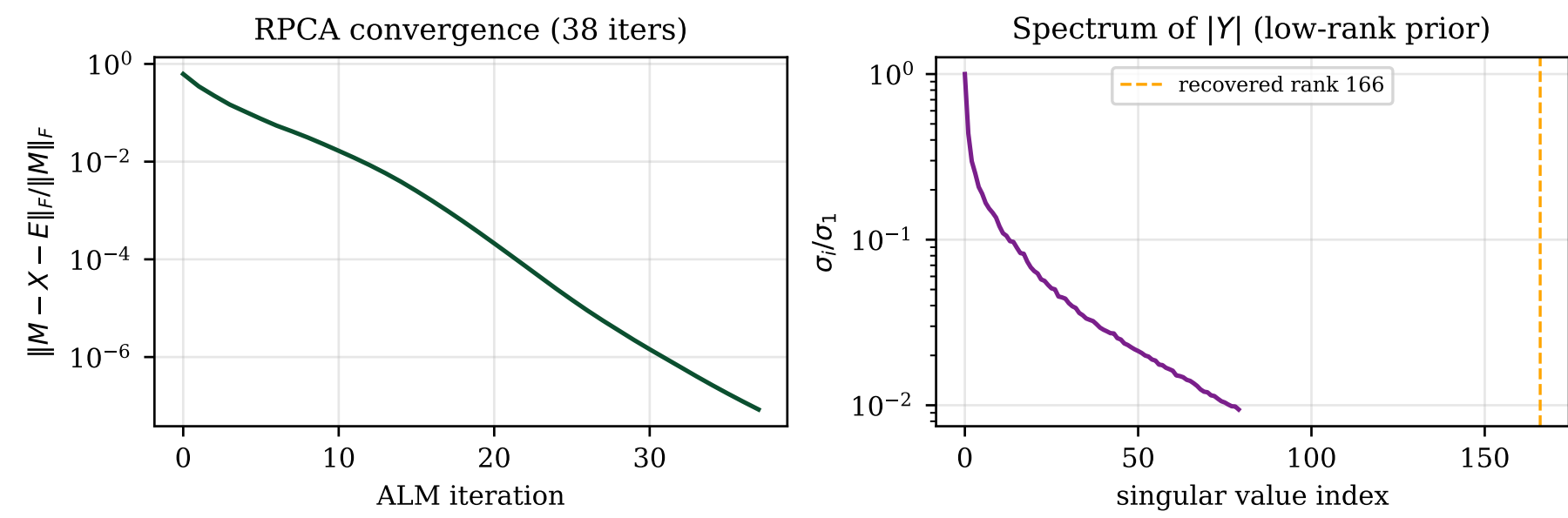


Mixture vs. vocals vs. accompaniment. The structure each stage exploits.

Stage A: Robust PCA (L13)

$$\min_{\mathbf{X}, \mathbf{E}} \|\mathbf{X}\|_* + \lambda \|\mathbf{E}\|_1 \quad \text{s.t.} \quad \mathbf{M} = \mathbf{X} + \mathbf{E}, \quad \lambda = \frac{1}{\sqrt{\max(F, T)}}$$

Low-rank $\mathbf{X}$ = accompaniment, sparse $\mathbf{E}$ = transients. Training-free, inexact ALM, converges to 10^{-7} in 38 iterations.



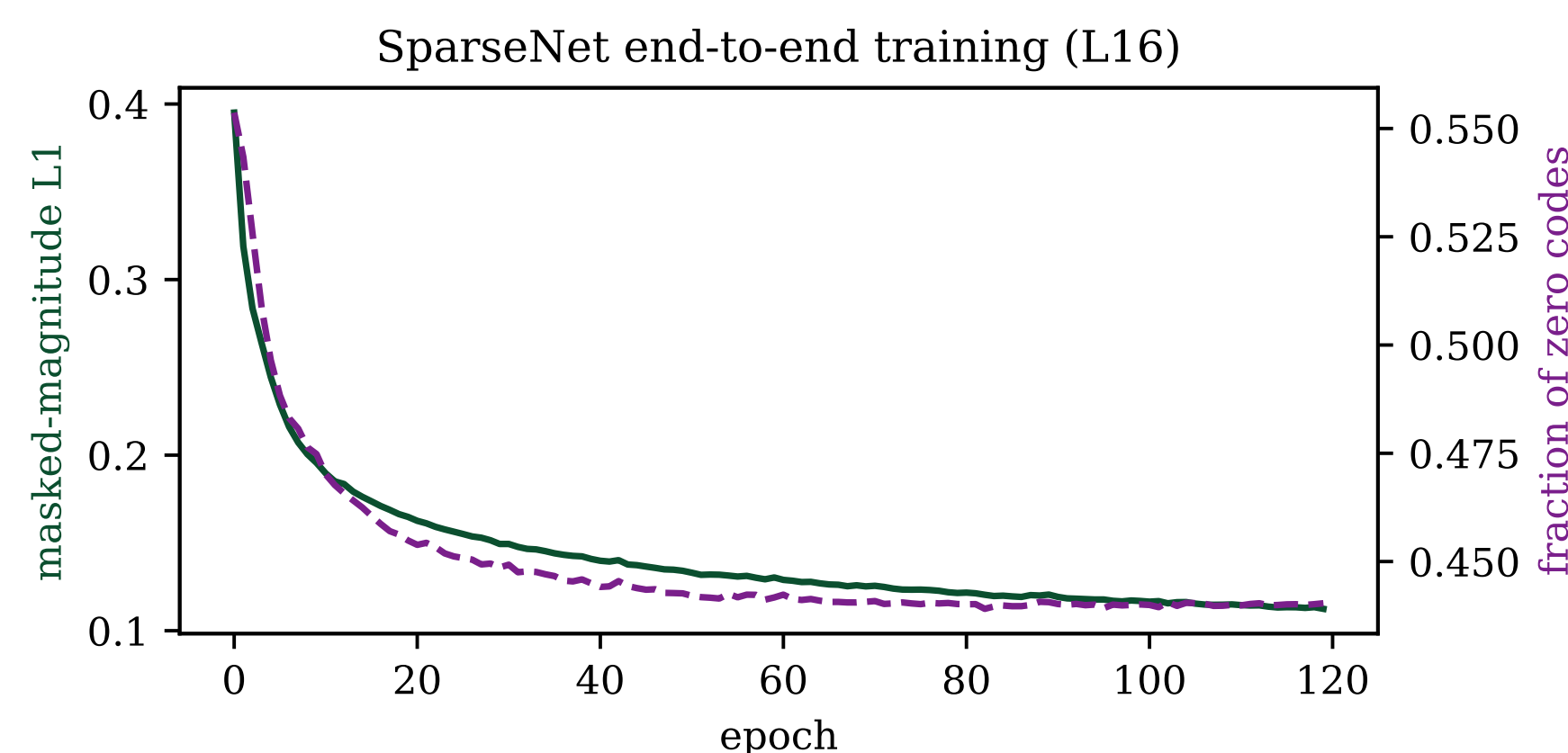
Stage D: SparseNet for spectrograms (L16)

The novel contribution. Chin’s SparseNet has only been tested on image classification. We adapt its **composite sparse-coding module** to **audio mask regression**. Each module maps a magnitude frame $\mathbf{x} \in \mathbb{R}^F$ through two non-negative sparse codes:

$$\boldsymbol{\alpha}^{(1)} = \arg \min_{\boldsymbol{\alpha} \geq 0} \frac{1}{2} \|\mathbf{x} - \mathbf{D}^{(1)} \boldsymbol{\alpha}\|_2^2 + \lambda_1 \|\boldsymbol{\alpha}\|_1 + \frac{\lambda_2}{2} \|\boldsymbol{\alpha}\|_2^2$$

$$\boldsymbol{\alpha}^{(2)} = \arg \min_{\boldsymbol{\alpha} \geq 0} \frac{1}{2} \|\boldsymbol{\alpha}^{(1)} - \mathbf{D}^{(2)} \boldsymbol{\alpha}\|_2^2 + \lambda_1 \|\boldsymbol{\alpha}\|_1 + \frac{\lambda_2}{2} \|\boldsymbol{\alpha}\|_2^2$$

with $\mathbf{D}^{(1)}$ *fat* (upsampling) and $\mathbf{D}^{(2)}$ *tall* (downsampling). We **unroll 25 steps of non-negative FISTA** per code and train 3 stacked modules **end-to-end by backprop** on Apple MPS. Dictionaries and shrinkage parameters are learned; the head emits a soft mask.



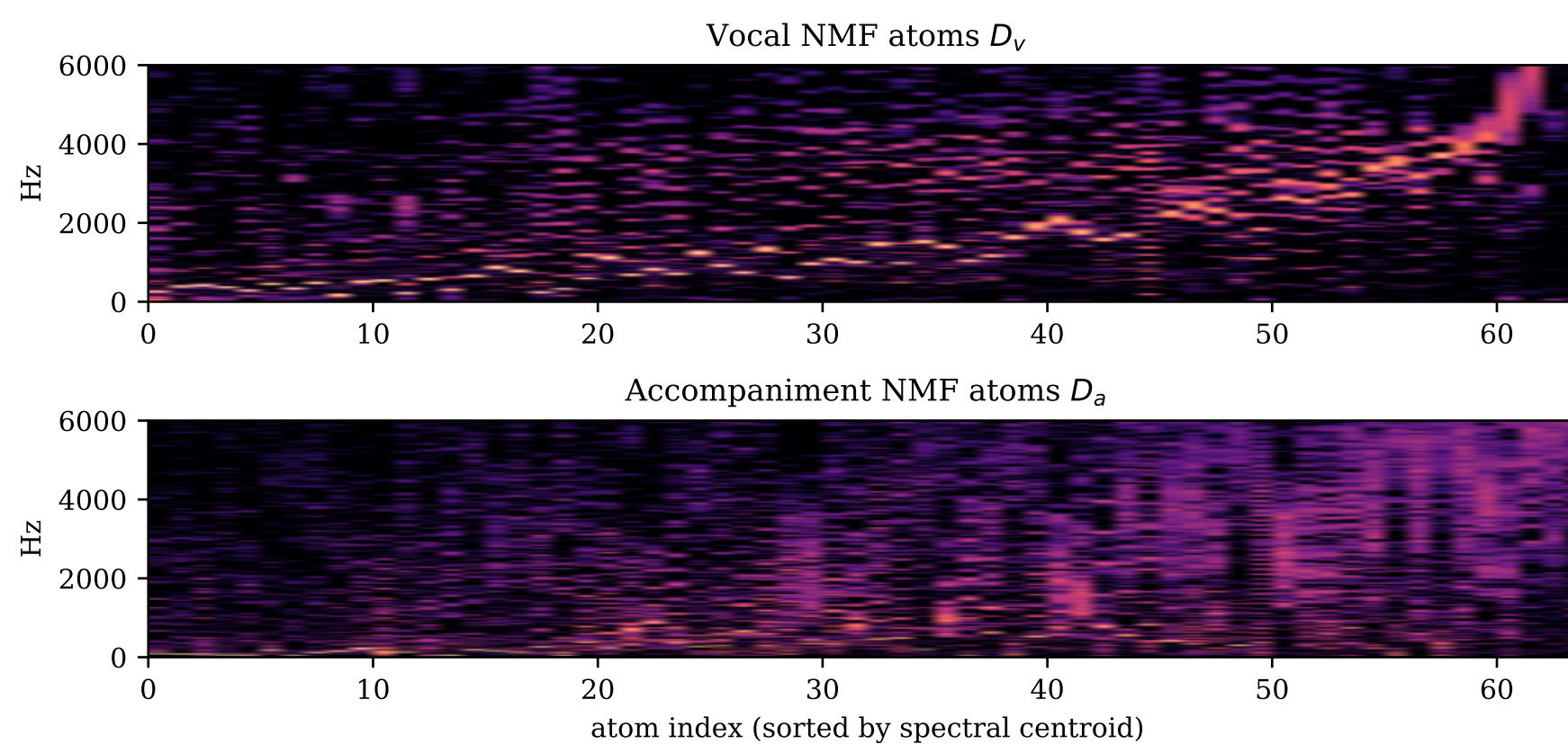
Codes stay $\sim 44\%$ sparse throughout end-to-end training.

The five-stage pipeline

Stage	Method	Lecture
A	Robust PCA (low-rank + sparse)	L13
B	Class-conditional NMF	L15
C	K-SVD dictionaries + SRC	L14, L8
D	SparseNet mask predictor	L16
E	MMV joint sparsity (stereo)	L12

Every stage reuses problem-set code: OMP (PS3), FISTA (PS5), SRC (PS7), nuclear-norm SVT (PS8).

Learned dictionaries (L14, L15)

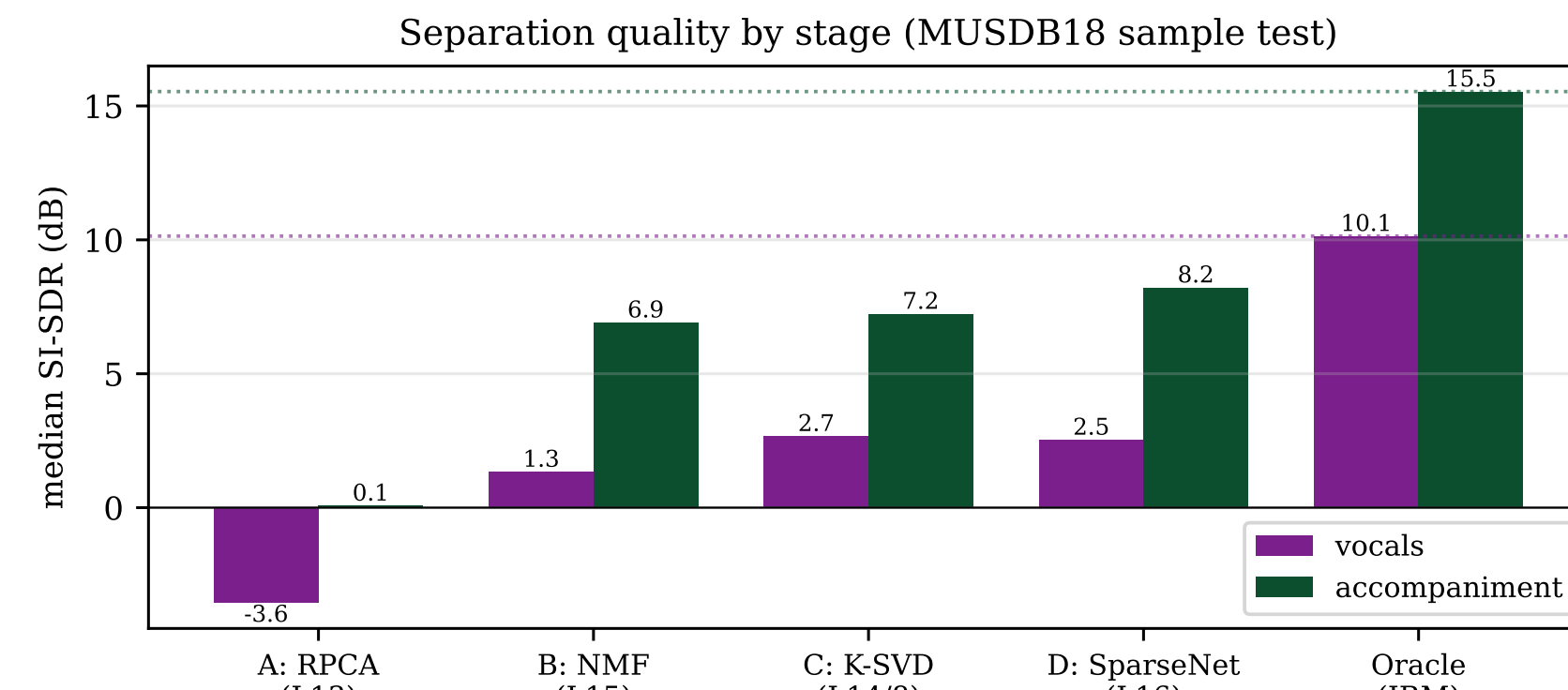


Vocal atoms ride the formant region; accompaniment atoms spread lower. Parts-based, exactly the L15 prediction.

Results: MUSDB18 sample

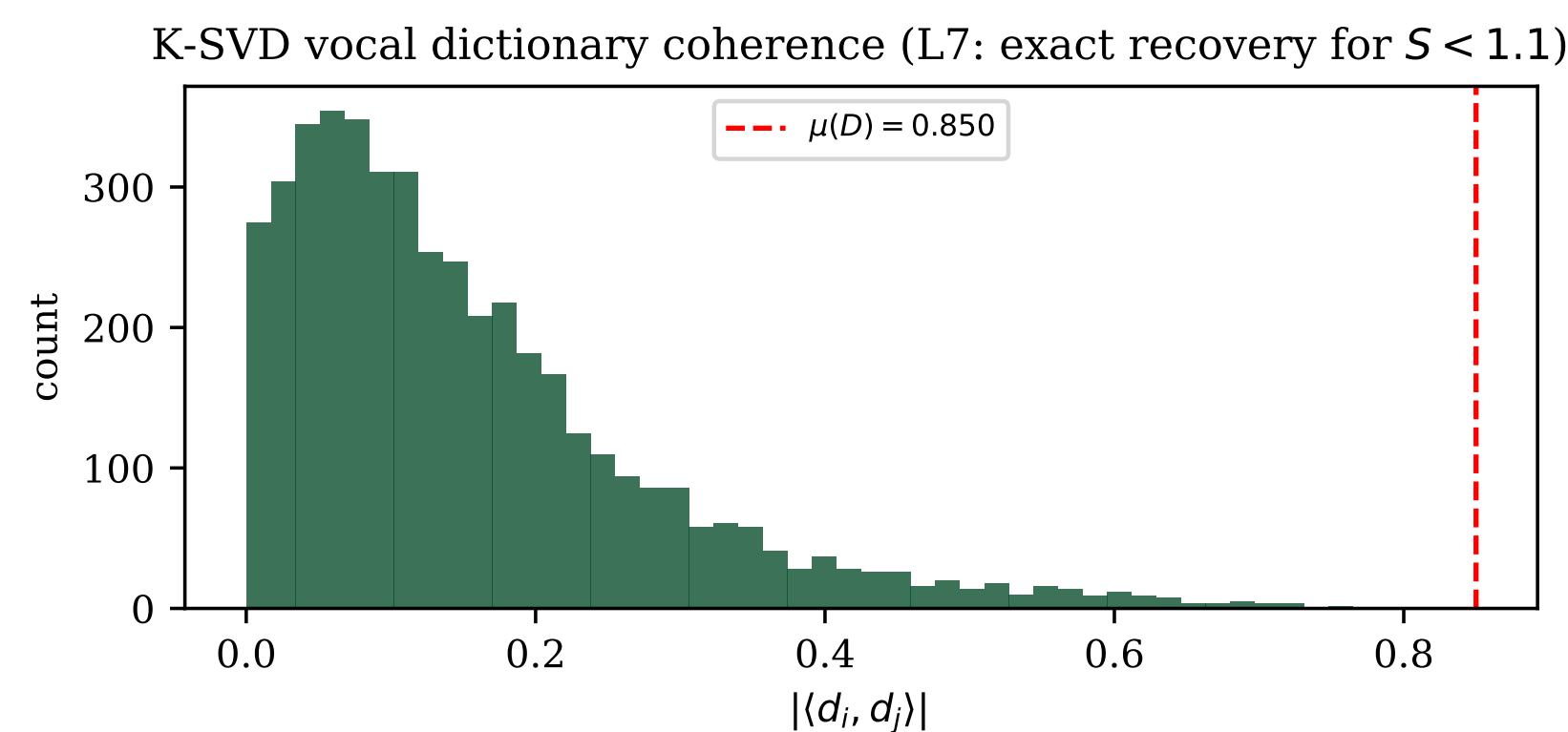
Stage	Lec.	Voc. Acc.
A. Robust PCA	L13	-3.6 0.1
B. NMF (KL)	L15	1.3 6.9
C. K-SVD + SRC L14/8		2.7 7.2
D. SparseNet	L16	2.5 8.2
Oracle (IRM)	—	10.1 15.5

Median SI-SDR (dB), 12 held-out tracks.



Clean unsupervised $\rightarrow$ **supervised** $\rightarrow$ **deep progression**. SparseNet matches discriminative K-SVD on vocals and leads on accompaniment.

An honest negative result



Learned K-SVD dictionaries have coherence $\mu \approx 0.85$. The L7 worst-case guarantee then certifies only $S \lesssim 1$, yet OMP at $S = 10$ works well. **Average-case** $\neq$ **worst-case** for trained dictionaries.

Takeaways

- Five course methods, one evaluable spectrogram pipeline.
- First adaptation of **SparseNet to audio mask regression** (L16).
- Theory (RIP, coherence, RPCA $\lambda = 1/\sqrt{N_1}$) connected to every stage, **including where it goes slack**.
- All code, trained weights, and **playable audio** released.
github.com/takakhoo/ENG109_FinalProject